



COMPETENCY STANDARD
FOR
ADVANCED WELDING 4G,5G & 6G
(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Advanced Welding 4G, 5G & 6G is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (27 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-AW-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	09
SICIP-SBD-AW-02-G	Carry Out Workplace Interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	09
SICIP-SBD-AW-03-G	Operate In a Team Environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	09
Total Hour			27 Hrs.

Sector Specific Competencies (45 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-AW-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	09
SICIP-SBD-AW-02-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	36
Total Hour			45 Hrs.

Occupation Specific Competencies (288 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-AW-01-O	Prepare Materials and Equipment for Advanced Welding	1. Select Base Materials and Consumables for 4G, 5G and 6G Welding 2. Arrange Welding Tools 3. Identify Welding Joints 4. Set up Welding Machines 5. Adjust Welding Parameters for Each Welding Position	18
SICIP- SBD-AW-02-O	Perform 4G Welding	1. Identify Workpieces for 4G Welding 2. Execute 4G Welding 3. Inspect Weld Bead 4. Clean and Prepare the Weld Area	81
SICIP- SBD-AW-03-O	Perform 5G Welding	1. Arrange Materials to 5G Welding Position 2. Execute 5G Welding 3. Inspect Weld Bead 4. Clean and Prepare the Weld Area	81
SICIP- SBD-AW-04-O	Perform 6G Welding	1. Identify Workpieces for 6G Welding 2. Execute 6G Welding 3. Inspect Weld Bead 4. Clean and Prepare the Weld Area	108
Total Hours			288 Hrs.

COMPETENCY STANDARD: ADVANCED WELDING 4G, 5G & 6G

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 09 hrs.	Unit Code: SICIP-SBD-AW-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations

2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, signs and symbols 4.3 Codes of conduct 4.4 Projector

	4.5 Learning manual 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 09 hrs.	Unit Code: SICIP-SBD-AW-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening

	1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	2.1 Demonstrating workplace communication and etiquette 2.2 Interpreting workplace instructions and symbols 2.3 Demonstrating active participation in workplace meeting 2.4 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 09 hrs.	Unit Code: SICIP-SBD-AW-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted.
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
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1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Workplace documents, Projector 4.3. Learning manual 4.4. Tools, equipment and facilities appropriate to the process or activities. 4.5. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration

	2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 09 hrs.	Unit Code: SICIP-SBD-AW-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during construction activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in construction sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair

	1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment 3.2 Adopting renewable energy sources 3.3 Implementing site protection measures 3.4 Using reusable materials 3.5 Choosing sustainable materials 3.6 Using noise-reducing equipment and scheduling work appropriately 3.7 Optimizing logistics and delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Asbestos waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in shipbuilding sector 1.2 Key terms and symbols related to environmental sustainability in construction drawings 1.3 Sources of environmental impacts in construction 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in construction activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of construction hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waste segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in construction sector 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK WITH HAND AND POWER TOOLS	Nominal Duration: 36 hrs.	Unit Code: SICIP-SBD-AW-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1. Appropriate <u>hand tools</u> are selected. 2.2. Safety precautions are ensured before using hand tools. 2.3. Unsafe or faulty hand tools are identified and marked for repair. 2.4. <u>Measuring tools</u> are checked and calibrated before use. 2.5. Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage. 4.4. Defective hand and power tools are inspected and

	repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
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2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
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2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: PREPARE MATERIALS AND EQUIPMENT FOR ADVANCED WELDING	Nominal Duration: 18 Hrs.	Unit Code: SICIP-SBD-AW-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to prepare materials and equipment for advanced welding. It specifically includes the task of selecting base materials and consumables for 4G, 5G and 6G welding, arranging welding tools, identifying welding joints, setting up welding machines and adjusting welding parameters for each welding positions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select base materials and consumables for 4G, 5G and 6G welding	1.1 Relevant <u>welding procedure specifications (WPS)</u> and <u>standards</u> are reviewed and interpreted. 1.2 <u>Base materials</u> are identified. 1.3 Welding consumables are selected based on base materials, welding position. 1.4 Welding consumables are inspected for defects.
2. Arrange welding tools	2.1 Required <u>welding tools</u> are identified. 2.2 Welding tools are inspected to ensure they are clean, functional, and free from defects. 2.3 Welding tools are arranged systematically in designated areas for easy accessibility during work. 2.4 Tools are stored securely to prevent damage and to maintain a safe working environment.
3. Identify welding joints	3.1 Safety precautions and <u>PPE</u> are used when inspecting and handling welding joints. 3.2 <u>Welding joint types</u> are identified. 3.3 Dimensions, angles, and fit-up of joints are identified.
4. Set up welding machines	4.1 Relevant welding machine manuals, specifications, and setup instructions are reviewed and understood. 4.2 The welding machine is inspected for any damage, wear, and safety compliance before setup. 4.3 Correct power supply is identified, and the welding machine is connected safely to the power source. 4.4 Required welding parameters are set according to the welding procedure specifications. 4.5 Electrodes are installed and fed correctly into the welding holder. 4.6 The welding machine is tested with trial runs to verify correct setup and functionality.
5. Adjust welding parameters for each	5.1 Welding positions and related welding specifications

welding positions	are understood. 5.2 Welding parameters such as current, voltage, travel speed, and polarity are identified according to the welding position. 5.3 Welding machine parameters are adjusted of each specific welding position. 5.4 Trial welds are conducted to ensure parameters produce acceptable weld quality in each position.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Welding procedure specifications (WPS)	1.1 Joint information 1.2 Technical information 1.3 Joint preparation 1.4 Welding details 1.5 Back gouging
2. Standards	2.1 AWS D1.1 2.2 ASME IX 2.3 ISO 5018
3. Base materials	3.1 4G: 3/8" or 1" plate 3.2 5G: fixed 6" schedule 40 pipe 3.3 6G: fixed 45° pipe, multi-pass all positions
4. Welding tools	4.1 Electrodes 4.2 Clamps 4.3 Grinders 4.4 Chipping hammers 4.5 Safety gear
5. PPE	5.1 Safety glasses 5.2 Ear plugs 5.3 Gloves 5.4 Apron 5.5 Helmet 5.6 Mask 5.7 Safety shoes
6. Welding joint types	6.1 Butt joints 6.2 Lap joints 6.3 Corner joints 6.4 T-joints 6.5 Thickness of parts 6.6 Root opening 6.7 Root face 6.8 Bevel angle 6.9 Groove angle 6.10 Effective throat thickness
7. Welding parameters	7.1 Current 7.2 Voltage

	7.3 Travel speed
	7.4 Polarity
	7.5 Number of passes

Curricular Content Guide

1. Underpinning Knowledge	1.1. Welding Procedure Specifications (WPS) and standards. 1.2. Identify base materials and material type and welding position 1.3. Identify necessary welding tools. 1.4. Identify welding joint types, dimensions, angles, and fit-up requirements. 1.5. Inspect the welding machine for safety, damage, or wear. 1.6. Identify and adjust parameters for each position.
2. Underpinning Skills	2.1. Identifying base materials 2.2. Selecting welding consumables based on base materials and welding position 2.3. Inspecting welding consumables for defects 2.4. Identifying required welding tools 2.5. Inspecting welding tools to ensure cleanliness, functionality, and freedom from defects 2.6. Arranging welding tools systematically for easy accessibility 2.7. Setting welding parameters according to welding procedure specifications 2.8. Installing and feeding electrodes correctly into welding holders 2.9. Testing welding machines with trial runs to verify setup and functionality 2.10. Storing tools securely to maintain safety and prevent damage
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 selected base materials and consumables for 4G, 5G and 6G welding 1.2 arranged welding tools 1.3 identified welding joints 1.4 set up welding machines 1.5 adjusted welding parameters for each welding positions
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM 4G WELDING	Nominal Duration: 81 Hrs.	Unit Code: SICIP- SBD-AW-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform 4G welding. It specifically includes the task of identifying workpieces for 4G welding, executing 4G welding, inspecting weld bead and cleaning and preparing the weld area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify workpieces for 4G welding	1.1 Safety precautions are followed and <u>PPEs</u> are used. 1.2 Welding procedure specifications (WPS) for 4G welding are reviewed and understood. 1.3 3/8" or 1" base plate are collected as per WPS. 1.4 Dimensions, thickness, and joint configuration of workpieces are measured and confirmed as per 4G welding. 1.5 Workpieces are marked and positioned in the overhead position.
2. Execute 4G welding	2.1 Appropriate welding machines and electrodes are prepared and checked for use. 2.2 Welding techniques including electrode angle, travel speed, and welding pattern are applied. 2.3 Heat input is controlled to prevent defects. 2.4 Weld pool and spatter is managed throughout the overhead weld and control against gravity. 2.5 Visual and tactile monitoring of the weld bead is conducted to ensure uniformity and absence of defects.
3. Inspect weld bead	3.1 Measurements of weld bead size, reinforcement, and profile are taken using appropriate gauges and tools. 3.2 Applicable <u>non-destructive testing (NDT) methods</u> are conducted to detect subsurface defects. 3.3 Any weld defects or irregularities are identified and classified according to severity. 3.4 The weld bead is verified against welding procedure specifications (WPS).
4. Clean and prepare the weld area	4.1 The weld area is inspected. 4.2 Appropriate cleaning and preparation methods are selected based on the weld area condition. 4.3 Contaminants and impurities are removed completely from the weld area using selected cleaning methods. 4.4 Prepared weld area is inspected to verify cleanliness and readiness for welding.

Range of Variables

Variable	Range (Includes but not limited to):
1. PPEs	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes
2. Non-destructive testing (NDT) methods	2.1 Dye penetrant 2.2 Magnetic particle 2.3 Ultrasonic testing 2.4 Radiograph

Curricular Content Guide

1. Underpinning Knowledge	1.1. Comprehend the welding procedure specifications (WPS) for 4G welding. 1.2. Understand welding machines and electrodes. 1.3. Measure weld bead size, reinforcement, and profile with appropriate gauges. 1.4. Non-destructive testing (NDT) methods to detect subsurface defects. 1.5. Identify and classify any defects or irregularities based on severity. 1.6. Cleaning and preparation methods based on the weld area condition.
2. Underpinning Skills	2.1. Reviewing and understanding welding procedure specifications (WPS) for 4G welding 2.2. Collecting base plates (3/8" or 1") as per WPS 2.3. Measuring and confirming dimensions, thickness, and joint configuration of workpieces 2.4. Marking and positioning workpieces in the overhead welding position 2.5. Preparing and checking welding machines and electrodes for use 2.6. Applying welding techniques including electrode angle, travel speed, and welding pattern 2.7. Controlling heat input to prevent weld defects 2.8. Managing weld pool and spatter during overhead welding, controlling against gravity 2.9. Conducting visual and tactile monitoring of the weld bead to ensure uniformity and absence of defects 2.10. Conducting non-destructive testing (NDT) methods to detect subsurface defects
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities

	3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified workpieces for 4G welding 1.2 executed 4G welding 1.3 inspected weld bead 1.4 cleaned and prepared the weld area
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM 5G WELDING	Nominal Duration: 81 Hrs.	Unit Code: SICIP-SBD-AW-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform 5G welding. It specifically includes the tasks of arranging materials to 5G welding position, executing 5G welding, inspecting weld bead and cleaning and preparing the weld area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Arrange materials to 5G welding position	1.1 Safety precautions are followed and <u>PPEs</u> are used. 1.2 Relevant welding procedure specifications (WPS) for 5G welding are understood. 1.3 Fixed 6" schedule 40 pipe are identified as per 5G welding requirement. 1.4 Workpieces are cleaned, beveled, and prepared as required. 1.5 Workpieces are arranged and clamped in the horizontal fixed position. 1.6 Alignment, gap, and fit-up between workpieces are checked and adjusted to meet welding standards.
2. Execute 5G welding	2.1 Safety protocols are followed throughout welding. 2.2 <u>Welding parameters</u> are set according to the WPS. 2.3 Welding beads are performed using proper electrode angle and travel speed. 2.4 Vertical up-and-down welding is carried out on a round surface. 2.5 <u>Electrode manipulation techniques</u> are applied. 2.6 Arc length and electrode angle are maintained consistently. 2.7 Weld bead is examined visually for surface defects such as cracks, porosity, undercut, and uniformity.
3. Inspect weld bead	3.1 Measurements of weld bead size, reinforcement, and profile are taken using appropriate gauges and tools. 3.2 Applicable <u>non-destructive testing (NDT) methods</u> are conducted to detect subsurface defects. 3.3 Any weld defects or irregularities are identified and classified according to severity. 3.4 The weld bead is verified against welding procedure specifications (WPS).

4. Clean and prepare the weld area	4.1	The weld area is inspected.
	4.2	Appropriate cleaning and preparation methods are selected based on the weld area condition.
	4.3	Contaminants and impurities are removed completely from the weld area using selected cleaning methods.
	4.4	Prepared weld area is inspected to verify cleanliness and readiness for welding.

Range of Variables

Variable	Range (Includes but not limited to):
1. PPEs	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes
2. Welding parameters	2.1 Current 2.2 Voltage 2.3 Travel speed
3. Electrode manipulation techniques	3.1 Weaving 3.2 Whipping 3.3 Circular motion
4. Non-destructive testing (NDT) methods	4.1 Dye penetrant 4.2 Magnetic particle 4.3 Ultrasonic testing 4.4 Radiograph

Curricular Content Guide

1. Underpinning Knowledge	1.1. Comprehend relevant welding procedure specifications (WPS) for 5G welding. 1.2. Welding parameters according to WPS. 1.3. Vertical up-and-down welding on a round surface. 1.4. Electrode manipulation techniques. 1.5. Correct electrode angle, travel speed, and proper welding techniques. 1.6. Weld bead for defects such as cracks, porosity, undercut, and uniformity. 1.7. Select appropriate cleaning and preparation methods based on weld area condition.
2. Underpinning Skills	2.1 Understanding relevant welding procedure specifications (WPS) for 5G welding 2.2 Identifying fixed 6" schedule 40 pipe as per 5G welding requirements 2.3 Cleaning, beveling, and preparing workpieces as required

	2.4 Arranging and clamping workpieces in the horizontal fixed position 2.5 Checking and adjusting alignment, gap, and fit-up between workpieces to meet welding standards 2.6 Following safety protocols throughout welding 2.7 Setting welding parameters according to the WPS 2.8 Performing welding beads using proper electrode angle and travel speed 2.9 Carrying out vertical up-and-down welding on a round surface 2.10 Conducting non-destructive testing (NDT) methods to detect subsurface defects
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 arranged materials to 5G welding position 1.2 executed 5G welding 1.3 inspected weld bead 1.4 cleaned and prepared the weld area
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM 6G WELDING	Nominal Duration: 108 Hrs.	Unit Code: SICIP-SBD-AW-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform 6G welding. It specifically includes the tasks of identifying workpieces for 6G welding, executing 6G welding, inspecting weld bead and cleaning and preparing the weld area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify workpieces for 6G welding	1.1 Workpieces are identified according to the 6G welding position. 1.2 Workpieces are positioned and clamped to prevent movement during welding. 1.3 Pipe ends are beveled accurately to ensure full penetration. 1.4 Uniform root gap (typically 1/8" and land/root face 1/16) are maintained. 1.5 Alignment of the workpieces is checked and adjusted to meet the required welding angles. 1.6 Preparation of the workpieces for welding is completed including cleaning and fit-up.
2. Execute 6G welding	2.1 Safety precautions are followed and <u>personal protective equipment (PPE)</u> is properly used. 2.2 Relevant welding procedure specifications (WPS) for 6G welding are understood. 2.3 Welding equipment and parameters are set according to the WPS. 2.4 <u>Welding techniques</u> are applied in accordance with the Welding Procedure Specification (WPS) 2.5 <u>Welding passes</u> are carried out as specified by the WPS to maintain quality and standards. 2.6 Welds are continuously monitored, and necessary adjustments are made to maintain weld integrity. 2.7 Welds are inspected using appropriate methods to verify compliance with applicable standards.
3. Inspect weld bead	3.1 Measurements of weld bead size, reinforcement, and profile are taken using appropriate gauges and tools. 3.2 Applicable <u>non-destructive testing (NDT) methods</u> are conducted to detect subsurface defects. 3.3 Any weld defects or irregularities are identified and classified according to severity. 3.4 The weld bead is verified against welding procedure specifications (WPS).

4. Clean and prepare the weld area	4.1 The weld area is inspected. 4.2 Appropriate cleaning and preparation methods are selected based on the weld area condition. 4.3 Contaminants and impurities are removed completely from the weld area using selected cleaning methods. 4.4 Prepared weld area is inspected to verify cleanliness and readiness for welding.
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Range of Variables

Variable	Range (Includes but not limited to):
1. PPEs	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes
2. Welding techniques	2.1 Body Positioning 2.1.1 Stable and comfortable stance 2.1.2 Weld from face or side at 45° angle 2.1.3 Ensure good visibility of weld pool 2.1.4 Keep steady arm support 2.2 Travel Angle 2.2.1 Work angle around 45° to pipe surface 2.2.2 Use drag (back) angle, 5° to 15° 2.2.3 Electrode angled slightly in travel direction 2.3 Arc Length 2.3.1 Maintain short arc (~1/8 inch)
3. Welding passes	3.1 Root Pass: Use 3/32" or 1/8" electrode or 7018 with backing. 3.2 Hot Pass: Remove slag and reinforce root; avoid undercut. 3.3 Fill & Cap Pass: Use slight weave or stringers per test specs. 3.4 Interpass: Clean with wire brush and chipping hammer; control interpass temperature.
4. Non-destructive testing (NDT) methods	4.1 Dye penetrant 4.2 Magnetic particle 4.3 Ultrasonic testing 4.4 Radiograph

Curricular Content Guide

1. Underpinning Knowledge	1.1. Safety precautions and use proper PPE throughout welding. 1.2. Identify workpieces according to the 6G welding
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	position. 1.3. Comprehend welding procedure specifications (WPS) for 6G welding. 1.4. Welding passes as specified in the WPS. 1.5. Measure weld bead size, reinforcement, and profile using correct tools and gauges. 1.6. Non-destructive testing (NDT) to detect any subsurface defects. 1.7. Inspect the weld area for contaminants and readiness.
2. Underpinning Skills	2.1. Identifying workpieces according to the 6G welding position 2.2. Positioning and clamping workpieces to prevent movement during welding 2.3. Beveling pipe ends to ensure full penetration 2.4. Maintaining uniform root gap and land/root face dimensions 2.5. Checking and adjusting alignment of workpieces to meet required welding angles 2.6. Preparing workpieces including cleaning and fit-up for welding 2.7. Setting welding equipment and parameters according to the WPS 2.8. Applying welding techniques in accordance with WPS 2.9. Carrying out welding passes as specified to maintain quality and standards 2.10. Conducting non-destructive testing (NDT) to detect subsurface defects
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified workpieces for 6G welding 1.2 executed 6G welding
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	1.3 inspected weld bead 1.4 cleaned and prepared the weld area
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Advanced Welding (4G,5G and 6G)
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	AC/DC Arc welding machine	10
5.	Gas welding set	03
6.	MIG welding machine	02
7.	TIG welding machine	02
8.	Gas cutting set	03
9.	Spare cylinder set (Argon, oxygen and acetylene)	01
10.	Power grinding machine	05
11.	Hand grinding machine	10
12.	Bench vice	10
13.	Tool box set for welding workshop	10
14.	Fire extinguisher	04
15.	Welding booth with exhaust system	10
16.	First aid box	03

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Advanced Welding (4G, 5G and 6G) the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.